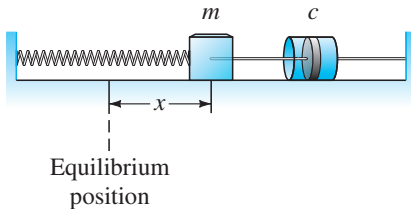


Section 3.4: A Model of Vibrations

Mass-Spring-Dashpot

- We consider a body of mass m attached to one end of an ordinary spring that resists compression as well as stretching.
- The other end of the spring is attached to a fixed wall, as shown in the figure.
- Assume that the body rests on a frictionless horizontal plane, so that it can move only back and forth as the spring compresses and stretches.



Section 3.4: A Model of Vibrations

Spring Force

- Denote by x the distance of the body from its **equilibrium position**—its position when the spring is unstretched.
- We take $x > 0$ when the spring is stretched, and thus $x < 0$ when it is compressed.
- According to Hooke's law, the restorative force F_S that the spring exerts on the mass is proportional to the distance x that the spring has been stretched or compressed.

$$F_S = -kx.$$

Section 3.4: A Model of Vibrations

Spring Force (cont'd)

- The positive constant of proportionality k is called the **spring constant**.
- Note that F_S and x have opposite signs: $F_S < 0$ when $x > 0$, $F_S > 0$ when $x < 0$.
- The dashpot is a device, like a shock absorber, that provides a force directed opposite to the instantaneous direction of motion of the mass m .

Section 3.4: A Model of Vibrations

Damping

- We assume the dashpot is so designed that this force F_R is proportional to the velocity $v = dx/dt$ of the mass; that is,

$$F_R = -cv = -c \frac{dx}{dt}.$$

- The positive constant c is the **damping constant** of the dashpot.
- More generally, we may regard this equation as specifying frictional forces in our system (including air resistance to the motion of m).

Section 3.4: A Model of Vibrations

External Force and Differential Equation

- If, in addition to the forces F_S and F_R , the mass is subjected to a given **external force** $F_E = F(t)$, then the total force acting on the mass is $F = F_S + F_R + F_E$.
- Using Newton's law

$$F = ma = m \frac{d^2x}{dt^2} = mx'',$$

we obtain the second-order linear differential equation

$$mx'' + cx' + kx = F(t)$$

that governs the motion of the mass.

Section 3.4: A Model of Vibrations

Undamped vs. Damped

- If there is no dashpot (and we ignore all frictional forces), then we set $c = 0$ in the differential equation

$$mx'' + cx' + kx = F(t)$$

and call the motion **undamped**.

- The motion is **damped** if $c > 0$.
- If there is no external force, we replace $F(t)$ with 0 in this differential equation.

Section 3.4: A Model of Vibrations

Free vs. Forced

- We refer to the motion as **free** in this case and **forced** in the case $F(t) \neq 0$.
- Thus the homogeneous equation

$$mx'' + cx' + kx = 0$$

describes free motion of a mass on a spring with dashpot but with no external forces applied.

Simple Harmonic Motion

- If we have only a mass on a spring, with neither damping nor external force, then our general differential equation

$$mx'' + cx' + kx = F(t)$$

takes the form

$$mx'' + kx = 0.$$

- It is convenient to define

$$\omega_0 = \sqrt{\frac{k}{m}}.$$

Simple Harmonic Motion

General Solution

- Then we can rewrite our equation in the form

$$x'' + \omega_0^2 x = 0.$$

Simple Harmonic Motion

Simple Harmonic Motion

Simple Harmonic Motion

Amplitude, Circular Frequency, and Phase Angle

- It has
 1. Amplitude C ,
 2. Circular frequency ω_0 , and
 3. Phase angle α .
- Such motion is called simple harmonic motion.
- If time t is measured in seconds, the circular frequency ω_0 has dimensions of radians per second (rad/s).

Simple Harmonic Motion

Period and Frequency

- The **period** of the motion is the time required for the system to complete one full oscillation, so is given by

$$T = \frac{2\pi}{\omega_0}$$

seconds.

- Its **frequency** is

$$\nu = \frac{1}{T} = \frac{\omega_0}{2\pi}$$

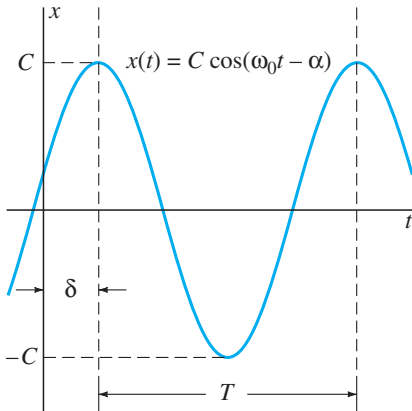
in hertz (Hz), which measures the number of complete cycles per second.

Simple Harmonic Motion

Typical Graph

- Here is a typical graph of a simple harmonic position function

$$\begin{aligned}x(t) &= C \cos(\omega_0 t - \alpha) \\&= C \cos\left(\omega_0 \left(t - \frac{\alpha}{\omega_0}\right)\right) \\&= C \cos(\omega_0(t - \delta)),\end{aligned}$$



showing the amplitude C , the period T , and the **time lag**

$$\delta = \frac{\alpha}{\omega_0}.$$

Differential Equation

- With damping but no external force, the differential equation we have been studying takes the form $mx'' + cx' + kx = 0$.
- Alternatively,

$$x'' + 2px' + \omega_0^2 x = 0,$$

where $\omega_0 = \sqrt{k/m}$ is the corresponding *undamped* circular frequency and

$$p = \frac{c}{2m} > 0.$$

Characteristic Roots

- The characteristic equation $r^2 + 2pr + \omega_0^2 = 0$ has roots

$$r_1, r_2 = -p \pm (p^2 - \omega_0^2)^{1/2}$$

that depend on the sign of

$$p^2 - \omega_0^2 = \frac{c^2}{4m^2} - \frac{k}{m} = \frac{c^2 - 4km}{4m^2}.$$

- The **critical damping** c_{cr} is given by $c_{\text{cr}} = \sqrt{4km}$, and we distinguish three cases, according as $c > c_{\text{cr}}$, $c = c_{\text{cr}}$, or $c < c_{\text{cr}}$.

Overdamped Case: $c > c_{\text{cr}}$ ($c^2 > 4km$)

- Because c is relatively large in this case, we are dealing with a strong resistance in comparison with a relatively weak spring or a small mass.
- Then

$$r_1, r_2 = -p \pm (p^2 - \omega_0^2)^{1/2}$$

gives distinct real roots r_1 and r_2 , both of which are negative.

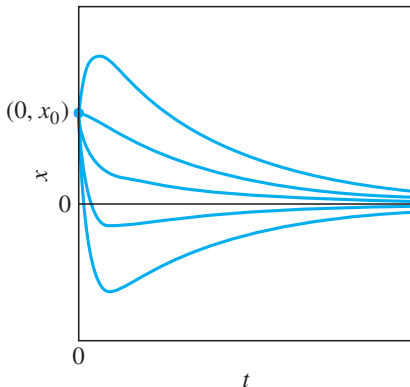
- The position function has the form

$$x(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}.$$

Types of Damping

Overdamped Case (cont'd)

- It is clear that $x(t) \rightarrow 0$ as $t \rightarrow +\infty$.
- The body settles to its equilibrium position without any oscillations.
- The figure shows some typical graphs of the position function.
- We chose x_0 a fixed positive number and illustrated the effects of changing the initial velocity v_0 .
- In every case the would-be oscillations are damped out.



Critically Damped Case: $c = c_{\text{cr}}$ ($c^2 = 4km$)

- In this case we have equal roots $r_1 = r_2 = -p$ of the characteristic equation, so the general solution is

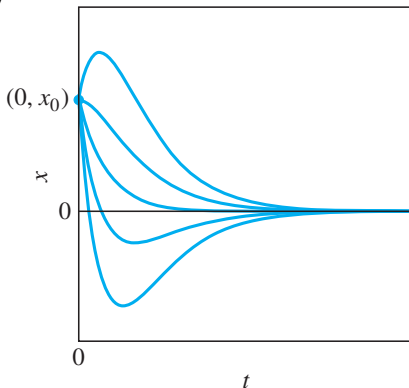
$$x(t) = e^{-pt}(c_1 + c_2t).$$

- Because $e^{-pt} > 0$ and $c_1 + c_2t$ has at most one positive zero, the body passes through its equilibrium position at most once.
- It is clear that $x(t) \rightarrow 0$ as $t \rightarrow +\infty$.
- Some graphs of the motion in the critically damped case are shown on the next slide.

Types of Damping

Critically Damped Case (cont'd)

- They resemble those of the overdamped case.
- In the critically damped case, the resistance of the dashpot is just large enough to damp out any oscillations.
- Yet even a slight reduction in resistance will bring us to the remaining case ...
- ... which is the case that shows the most dramatic behavior.



Types of Damping

Underdamped Case: $c < c_{cr}$ ($c^2 < 4km$)

- The characteristic equation now has two complex conjugate roots $-p \pm i\sqrt{\omega_0^2 - p^2}$, and the general solution is

$$x(t) = e^{-pt} (A \cos \omega_1 t + B \sin \omega_1 t),$$

where

$$\omega_1 = \sqrt{\omega_0^2 - p^2} = \frac{\sqrt{4km - c^2}}{2m}.$$

- Using the cosine addition formula we may rewrite $x(t)$ as

$$x(t) = Ce^{-pt} \left(\frac{A}{C} \cos \omega_1 t + \frac{B}{C} \sin \omega_1 t \right).$$

Underdamped Case (cont'd)

- This leads to

$$x(t) = Ce^{-pt} \cos(\omega_1 t - \alpha)$$

where

$$C = \sqrt{A^2 + B^2}, \quad \cos \alpha = \frac{A}{C}, \quad \text{and} \quad \sin \alpha = \frac{B}{C}.$$

- This solution represents exponentially damped oscillations of the body around its equilibrium position.
- The graph of $x(t)$ lies between the “amplitude envelope” curves $x = -Ce^{-pt}$ and $x = Ce^{-pt}$ and touches them when $\omega_1 t - \alpha$ is an integral multiple of π .

Types of Damping

Underdamped Case (cont'd)

- The motion is not actually periodic.
- Yet it is useful to call ω_1 its **circular frequency** (more properly, its **pseudofrequency**).
- Also $T_1 = 2\pi/\omega_1$ is its **pseudoperiod** of oscillation.
- Finally, Ce^{-pt} is its **time-varying amplitude**.

